

CHL5233: Quarto in-class assignment

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For this in-class assignment, you will work in a group to create your **individual** Quarto document. When finished, render the document to PDF, proofread it, and commit and push the edited file and new PDF to the new repository.

Create a new repository

Create a GitHub repository called `armed_conflict`. Create the folders `data` and `reports`; save `countries.txt` and `regions.txt` in `data/raw/`; and save `quarto_inclass.qmd` in `reports/`.

Edit this file to create a PDF document using Quarto

The setup chunk is configured to run while hiding its code, results, warnings, and messages from the final PDF.

First

Read in `countries.txt` and display the number of rows and columns.

```
countries <- read.table(  
  "../data/raw/countries.txt",  
  header = TRUE  
)
```

```
dim(countries)
```

```
[1] 186 2
```

There are 186 countries included in the `countries` data set.

Second

Read in `regions.txt` and display the number of rows and columns.

```
regions <- read.table(  
  "../data/raw/regions.txt",  
  header = TRUE,  
  fileEncoding = "latin1"  
)
```

```
dim(regions)
```

```
[1] 249 6
```

There are 249 countries and areas included in the `regions` data set.

Third

Subset `regions` to only include countries in `countries`, matching on the three-digit ISO country code.

```
regions_subset <- regions %>%  
  filter(country_code %in% countries$ISO)
```

```
dim(regions_subset)
```

```
[1] 186 6
```

After matching by ISO country code, all 186 countries are retained.

Tabulate countries by region and sub-region

Use `group_by()` and `summarise()` to calculate counts, then display them with `kable()`.

```
region_table <- regions_subset %>%  
  group_by(region) %>%  
  summarise(Count = n())
```

```
kable(  
  region_table,  
  col.names = c("Region", "Count")  
)
```

Region	Count
Africa	54
Americas	35
Asia	46
Europe	40
Oceania	11

```
subregion_table <- regions_subset %>%  
  group_by(sub_region) %>%  
  summarise(Count = n()) %>%  
  arrange(desc(Count))
```

```
kable(  
  subregion_table,  
  col.names = c("Sub-region", "Count")  
)
```

Sub-region	Count
Sub-Saharan Africa	48
Latin America and the Caribbean	33
Western Asia	16
Southern Europe	13
South-eastern Asia	11
Eastern Europe	10
Northern Europe	10
Southern Asia	9
Western Europe	7
Northern Africa	6
Central Asia	5
Eastern Asia	5
Melanesia	4
Micronesia	3
Australia and New Zealand	2
Northern America	2
Polynesia	2

Data visualization using ggplot

The figure is colour-coded by region, uses horizontal bars, and is ordered from the sub-region with the most countries at the top to the fewest at the bottom.

```
plot_data <- regions_subset %>%
  mutate(
    sub_region = forcats::fct_rev(
      forcats::fct_infreq(sub_region)
    )
  )

ggplot(
  plot_data,
  aes(x = sub_region, fill = region)
) +
  geom_bar() +
  coord_flip() +
  labs(
    x = "Sub-region",
    y = "Count",
    fill = "Region"
  ) +
  theme_minimal()
```

